

TPL-05 · FREE TEMPLATES

Progress measurement and rules of credit sheet

Five techniques, one rule-of-credit library, and a roll-up that weights by budget rather than averaging

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

Per cent complete is an assertion, and the sheet that produces it is where the assertion is either evidenced or invented. This template gives the measurement sheet and the rule-of-credit library behind it: five techniques with the arithmetic for each, a defined credit step for every claim, the evidence required before credit is released, and a roll-up that weights activities by budget instead of averaging their percentages.

— About this document

Document	TPL-05
Series	Free Templates
Version	1.0
Status	draft
Date	2026-08-04
Unresolved placeholders	0

Notice. This document is an educational publication of Project Controls Institute Global, Inc., and does not constitute accounting, tax, legal, financial or other professional advice. It must not be relied upon as such. Where a topic depends on the contract or the governing law, entitlement and treatment vary by jurisdiction; take local advice.

Standards. Standards and frameworks are referred to by name and described in this publication's own words. No standard's text, tables, diagrams or question banks are reproduced. All trademarks remain the property of their respective owners, and no endorsement by any standards body, employer or vendor is implied or should be inferred.

Status. Project Controls Institute Global, Inc. is a Delaware Non-Stock Corporation. The Institute has been developed with reference to ISO/IEC 17024 personnel-certification principles and is **not accredited** by ANAB, IAS or any other accreditation body. It intends to seek 501(c)(3) recognition, which has not been granted. No governmental approval, academic equivalence, or guaranteed employment, promotion or salary outcome is claimed or implied.

Examples. All worked examples are illustrative and fictional. Figures are chosen to demonstrate a method, not to describe any real project, organisation or market. Sample examination items, where they appear, are study material maintained separately from any live examination bank.

Completeness. Passages marked **[CONFIRM: ...]** are operational or market facts that have not yet been confirmed from a cited source. A document carrying them is a working draft and is not approved for publication.

© 2026 Project Controls Institute Global, Inc. · All rights reserved.

Contents

1. When to use this	4
2. How to complete it.....	4
3. The template	6
3.1 Sheet 1 — the measurement sheet.....	6
3.2 Sheet 2 — the rule-of-credit library.....	8
4. Worked fragment.....	9
4.1 The measurement sheet	9
4.2 The rules of credit behind the two milestone rows	10
4.3 What averaging would have cost	11
5. Common mistakes	11
6. Adapting it.....	12
7. Completion checklist	12
Related	12
Sources and standards.....	13
Status and version	13

Progress measurement and rules of credit sheet

In one paragraph. Per cent complete is an assertion, and the sheet that produces it is where the assertion is either evidenced or invented. This template gives the measurement sheet and the rule-of-credit library behind it: five techniques with the arithmetic for each, a defined credit step for every claim, the evidence required before credit is released, and a roll-up that weights activities by budget instead of averaging their percentages.

Who this is for. Cost engineers and planners who produce the monthly progress position; control account managers who claim it; and project controls managers who have to defend it when the earned value it produces is challenged.

— 1. When to use this

Build the sheet before the first progress claim, not after the first argument. The measurement technique for an activity is a decision that belongs with the baseline: it is recorded in the work breakdown structure dictionary (TPL-02) when the element is created, and the rule of credit is approved before any credit is claimed against it.

Use it every reporting period, at the cut-off defined in the controls execution plan (TPL-01 §3.6). Its output feeds three places:

- **Earned value**, where per cent complete multiplied by the budget at completion of the element gives the earned value used in TPL-07.
- **The monthly report**, where period movement is the number that tells a reader whether the project is accelerating or stalling.
- **Payment**, on projects where interim application is driven by measured progress. Where that is the case, say so on the sheet, because it changes the incentives on every claim.

The one time not to use it is on an activity whose output is genuinely unmeasurable and whose budget is material. There is no technique that rescues that; the answer is to decompose the activity until part of it becomes measurable, or to accept level of effort and state plainly that the element earns to plan.

— 2. How to complete it

Choose the technique from the work, not from convenience. The five techniques, and the condition each requires:

Technique	Use when	Per cent complete is
Units complete	Output is countable and homogeneous, and the total quantity is known	Quantity installed ÷ budgeted quantity
Incremental milestone	One work item passes through a fixed sequence of steps	The cumulative credit of the steps completed, in order
Weighted milestone	A deliverable is reached through discrete events that may complete out of order	The sum of the credits of the milestones achieved
Level of effort	The work is support or supervision, with no output of its own	Elapsed duration ÷ planned duration
Apportioned effort	The work varies directly with another activity and has no independent output	The per cent complete of the named base activity

Distinguish incremental from weighted milestone properly. Both assign credit at defined points. The difference is whether out-of-order achievement is permitted. Incremental milestone is a sequence: a step cannot be credited before its predecessor, and the sheet should refuse it. Weighted milestone is a set: each milestone carries a weight and may be achieved in any order. Choosing incremental where the work is genuinely non-sequential forces claimants to under-report; choosing weighted where the work is sequential lets them claim the easy milestones first.

Level of effort must be capped and must be small. Level of effort earns to plan by construction, so it generates no cost variance from schedule causes and never reports a schedule problem. Two disciplines follow: cap it at 100 per cent so it cannot keep earning past its planned finish, and keep it to a small share of the control account. If most of a control account is level of effort, its cost performance index is measuring almost nothing.

Apportioned effort must name its base activity. Not a discipline, not a control account — one activity, recorded on the sheet. Its per cent complete is then a formula referencing that row, so it cannot drift.

Write the rules of credit before the first claim, and freeze them. Changing a credit step mid-project changes reported progress without any work being done. If a rule must change, treat it as a measurement basis change under [TPL-04](#) and report the effect separately for one period.

Require objective evidence at every credit step. The evidence reference on the sheet is a document number, a survey, a test certificate, an inspection release — not a name. "Confirmed by the site manager" is not evidence; it is the same assertion again.

Roll up by weighting, never by averaging. The rolled-up per cent complete of a control account is total earned budget hours divided by total budget hours, not the mean of the activity percentages. This is the single most common arithmetic error in the discipline and §4.3 shows what it costs.

Store percentages as decimal fractions with percentage formatting. Type [0.55](#) into a cell formatted as a percentage, not [55](#). A sheet that mixes the two produces credit totals of 5,500 per cent and roll-ups that are silently wrong by a factor of a hundred.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

— 3. The template

3.1 Sheet 1 — the measurement sheet

Col	Field	Type	Definition and entry rule
A	Activity ID	Text	Matches the schedule activity identifier
B	Activity description	Text	
C	WBS element / control account	Text	From TPL-02
D	Technique	List	UC units complete · IM incremental milestone · WM weighted milestone · LOE level of effort · AE apportioned effort
E	Rule of credit reference	Text	The rule identifier in Sheet 2. Required for IM and WM .
F	Base activity (apportioned effort only)	Text	The activity ID whose per cent complete this follows
G	Unit of measure	Text	For UC — piles, cubic metres, welds, drawings, tonnes
H	Budgeted quantity	Number	For UC — the total scope in the stated unit
I	Budgeted hours	Number	The weight used in the roll-up. Cost may be used instead, but not both in one sheet.
J	Quantity to date / credit basis	Number	UC : quantity installed. IM/WM : cumulative credit from Sheet 2. LOE : elapsed duration. AE : leave blank.
K	Planned duration (LOE only)	Number	Same unit as column J for level-of-effort rows
L	Per cent complete this period	Calculated	See below
M	Per cent complete last period	Number	Carried forward from the previous issue of the sheet
N	Period movement	Calculated	
O	Earned quantity	Calculated	For UC rows — a cross-check
P	Earned hours	Calculated	The figure the roll-up uses
Q	Evidence reference	Text	Document, survey, certificate or inspection release number
R	Claimed by / verified by	Text	Two names, not one
S	Exception flag	Calculated	

Calculated column L — Per cent complete. In words, by technique:

- **Units complete** — quantity installed divided by budgeted quantity.
- **Incremental and weighted milestone** — the cumulative credit of the milestones achieved, brought in from Sheet 2.
- **Level of effort** — elapsed duration divided by planned duration, capped at 100 per cent.
- **Apportioned effort** — the per cent complete of the named base activity.

Spreadsheet, as one expression covering all five:

```
=IF($D2="UC",IF($H2=0,"",$J2/$H2),
  IF(OR($D2="IM",$D2="WM"),IF($J2="","", $J2),
  IF($D2="LOE",IF($K2=0,"",MIN(1,$J2/$K2)),
  IF($D2="AE",IFERROR(INDEX($L$2:$L$500,MATCH($F2,$A$2:$A$500,0)),""),""))))
```

Every division is guarded against a zero denominator. The apportioned-effort branch looks up the base activity's per cent complete by activity identifier; the **IFERROR** returns blank when the base activity cannot be found, which is itself a defect worth seeing. If you prefer to keep the branches separate, split the sheet by technique — the arithmetic is identical.

Calculated column N — Period movement. In words: this period's per cent complete less last period's, expressed in percentage points.

```
=IF(OR($L2="",$M2=""), "", $L2-$M2)
```

Calculated column O — Earned quantity. In words: per cent complete multiplied by the budgeted quantity. For a units-complete row this must equal the quantity to date, which makes it a free check that the technique and the entry agree.

```
=IF(OR($L2="",$H2=""), "", $L2*$H2)
```

Calculated column P — Earned hours. In words: per cent complete multiplied by the budgeted hours.

```
=IF(OR($L2="",$I2=""), "", $L2*$I2)
```

Calculated column S — Exception flag. In words: flag a per cent complete above 100 per cent, a backwards movement, or a milestone row with no rule reference.

```
=IF($L2="","",IF($L2>1,"OVER 100 % – quantity growth: raise a change or re-baseline the quantity",
  IF($N2<0,"NEGATIVE MOVEMENT – explain",
  IF(AND(OR($D2="IM",$D2="WM"),$E2=""),"NO RULE OF CREDIT",""))))
```

Do not cap a units-complete row at 100 per cent. A quantity that has exceeded its budget is real information — it means the take-off was wrong or the scope has grown — and capping it hides the very thing the sheet should surface. Level of effort is different and is capped, because it has no quantity to grow.

Roll-up to a control account. In words: total earned hours divided by total budgeted hours across the activities in the account. Spreadsheet, over the sheet's data range:

```
=IF(SUMIF($C$2:$C$500,$C2,$I$2:$I$500)=0,"",SUMIF($C$2:$C$500,$C2,$P$2:$P$500)/SUM-
  IF($C$2:$C$500,$C2,$I$2:$I$500))
```

Earned value from the roll-up. In words: the control account's rolled-up per cent complete multiplied by its budget at completion. That figure is the earned value input to **TPL-07**, whose treatment of the resulting measures this document does not repeat.

3.2 Sheet 2 — the rule-of-credit library

One block per rule. Rules are shared across activities of the same type; that is the point of a library.

Col	Field	Type	Definition
A	Rule ID	Text	e.g. ROC-CIV-03
B	Technique	List	IM or WM
C	Applies to	Text	The work type this rule measures
D	Step / milestone	Text	The event that releases credit
E	Sequence number	Number	For IM, the order that must be observed. Blank for WM.
F	Credit	Number	The credit released by this step, as a decimal fraction
G	Cumulative credit	Calculated	For IM only
H	Objective evidence required	Text	The document or test that proves the step
I	Released by (role)	Text	Who confirms the evidence
J	Achieved this period	List	Yes / No
K	Date achieved	Date	
L	Approved by / date	Text	Approval of the rule itself, not of a claim

Calculated column G — Cumulative credit. In words: the running total of credit down the sequence.

```
=IF($E2="", "", SUMIFS($F$2:F2, $A$2:A2, $A2))
```

Rule total validation. In words: the credits within a rule must total exactly 100 per cent.

```
=IF(ROUND(SUMIF($A$2:$A$200, $A2, $F$2:$F$200), 6)=1, "OK", "CHECK – credits must total 100 %")
```

ROUND to six decimal places prevents a false failure from floating-point representation, which is otherwise a genuine nuisance on rules with thirds.

Claimed credit for a rule. In words: for weighted milestone, the sum of the credits of milestones marked achieved. For incremental milestone, the cumulative credit at the highest sequence number achieved, which enforces the ordering.

Weighted milestone:

```
=SUMIFS($F$2:$F$200, $A$2:$A$200, $A2, $J$2:$J$200, "Yes")
```

Incremental milestone:

```
=IF(COUNTIFS($A$2:$A$200, $A2, $J$2:$J$200, "Yes")=0, 0, SUMIFS($F$2:$F$200, $A$2:$A$200, $A2, $E$2:$E$200, "<=" & MAXIFS($E$2:$E$200, $J$2:$J$200, "Yes")))
```

The incremental expression deliberately credits everything up to the highest achieved step, so a claim that skips a step credits the skipped step too — which is the correct behaviour for a

sequence, and makes an out-of-order claim visible rather than profitable. Where **MAXIFS** is unavailable, sort the rule by sequence and use a lookup on the last **Yes**.

4. Worked fragment

Illustrative figures. Control account CA-1000, civil works, on a fictional facility upgrade project. The period is May 2026, cut-off 31 May 2026. The weighting basis is budgeted direct labour hours. Currency is not used on this sheet; the resulting earned value is expressed in generic currency units (CU) thousands in **TPL-07**. Percentages are shown to one decimal place, rounded half away from zero; hours to whole hours.

4.1 The measurement sheet

Activity	Description	Tech.	Rule	Unit	Budget qty	Budget hrs	Qty / basis	% this period	% last period	Movement (pp)	Earn hrs
ACT-1010	Bulk excavation	UC	—	m ³	12,000	2,000	9,600	80.0 %	68.0 %	+12.0	1,600
ACT-1020	Bored piling	UC	—	piles	240	7,800	108	45.0 %	37.5 %	+7.5	3,510
ACT-1030	Pile caps and ground beams	IM	ROC-CIV-03	—	—	1,200	50.0 %	50.0 %	25.0 %	+25.0	600
ACT-1040	Structural steel erection	WM	ROC-STR-01	—	—	6,000	60.0 %	60.0 %	20.0 %	+40.0	3,600
ACT-1090	Civil supervision	LOE	—	months	—	2,000	11 of 20	55.0 %	50.0 %	+5.0	1,100
ACT-1095	Steel inspection and quality assurance	AE	base ACT-1040	—	—	800	—	60.0 %	20.0 %	+40.0	480
Control account CA-1000						19,800		55.0 %	35.1 %	+19.9	10,890

Verification of the activity arithmetic. Excavation: $9,600 \div 12,000 = 0.800$; earned hours $0.800 \times 2,000 = 1,600$. Piling: $108 \div 240 = 0.450$; earned hours $0.450 \times 7,800 = 3,510$. Pile caps: cumulative credit 50.0 per cent from rule **ROC-CIV-03** below; earned hours $0.500 \times 1,200 = 600$. Steel: achieved credit 60.0 per cent from rule **ROC-STR-01** below; earned hours $0.600 \times 6,000 = 3,600$. Supervision: $11 \div 20 = 0.550$; earned hours $0.550 \times 2,000 = 1,100$. Inspection: follows ACT-1040 at 60.0 per cent; earned hours $0.600 \times 800 = 480$.

Verification of the roll-up. Budgeted hours: $2,000 + 7,800 + 1,200 + 6,000 + 2,000 + 800 = 19,800$. Earned hours: $1,600 + 3,510 + 600 + 3,600 + 1,100 + 480 = 10,890$. Per cent complete = $10,890 \div 19,800 = 0.550 = 55.0 \text{ per cent}$. Last period's earned hours were $1,360 + 2,925 + 300 + 1,200 + 1,000 + 160 = 6,945$, giving $6,945 \div 19,800 = 0.3508 = 35.1 \text{ per cent}$, so the movement is 19.9 percentage points.

Earned value. Control account CA-1000 has a budget at completion of CU 4,000 thousand (TPL-02 §4.1). Earned value = $0.550 \times 4,000 = \text{CU } 2,200 \text{ thousand}$, which is the earned value entered for CA-1000 in TPL-07 §4.

A feature, not a fault. The 40-point movement on ACT-1040 is what milestone techniques do: nothing was credited while the frame was being erected, and 40 per cent landed the day the primary frame was signed off. Milestone progress is lumpy by construction. If a reader needs a smoother curve, the answer is more milestones with smaller weights, not a technique that credits partial achievement of a milestone.

4.2 The rules of credit behind the two milestone rows

ROC-CIV-03 — pile caps and ground beams, incremental milestone. Sequential; a step cannot be credited before its predecessor.

Seq	Step	Credit	Cumulative	Objective evidence	Achieved
1	Setting out complete and checked	10.0 %	10.0 %	Setting-out survey record	Yes
2	Blinding placed	15.0 %	25.0 %	Pour record	Yes
3	Reinforcement fixed and inspected	25.0 %	50.0 %	Reinforcement inspection release	Yes
4	Concrete placed	40.0 %	90.0 %	Pour record and delivery tickets	No
5	Stripped, cured and survey accepted	10.0 %	100.0 %	28-day cube results and as-built survey	No
Total		100.0 %			

Credit claimed: cumulative credit at the highest achieved sequence, step 3 → **50.0 per cent**. Verification: $10.0 + 15.0 + 25.0 = 50.0$.

ROC-STR-01 — structural steel erection, weighted milestone. Milestones may be achieved in any order; credit is the sum of those achieved.

Milestone	Credit	Objective evidence	Achieved
Steel delivered to site and receipt-inspected	20.0 %	Delivery notes and receipt inspection record	Yes
Primary frame erected and bolted	40.0 %	Erection sign-off sheet	Yes
Secondary steel erected	15.0 %	Erection sign-off sheet	No
Frame aligned, plumbed and surveyed	15.0 %	Survey report	No
Grouted and final inspection released	10.0 %	Inspection release	No
100.0 %			

Credit claimed: $20.0 + 40.0 = \text{60.0 per cent}$. Both rule totals return **OK** on the validation formula.

4.3 What averaging would have cost

The six activity percentages are 80.0, 45.0, 50.0, 60.0, 55.0 and 60.0. Their simple average is $350.0 \div 6 = 58.3$ per cent. The correct budget-weighted figure is 55.0 per cent. Applied to the control account's budget at completion of CU 4,000 thousand, the average would report earned value of $0.5833 \times 4,000 = \text{CU } 2,333$ thousand against the correct CU 2,200 thousand — **CU 133 thousand of earned value that has not been earned**, and a cost performance index overstated by the same margin.

The error is small here only because the budgets are not wildly unequal. Take two activities: one of 100 budget hours at 90 per cent complete, and one of 9,900 budget hours at 10 per cent. The simple average is $(90 + 10) \div 2 = 50.0$ per cent. The weighted answer is $(0.90 \times 100 + 0.10 \times 9,900) \div 10,000 = (90 + 990) \div 10,000 = \mathbf{10.8 \text{ per cent}}$. The averaging method reports a project nearly half complete that has barely started.

— 5. Common mistakes

Averaging activity percentages. See §4.3. It is the error that survives longest because the answer always looks plausible.

Level of effort used because measurement is inconvenient. Level of effort earns to plan and therefore never reports a problem. A control account that is mostly level of effort has a cost performance index that measures the passage of time.

Level of effort with no cap. Past its planned finish it keeps earning and eventually reports more than 100 per cent complete, which then propagates into earned value greater than budget at completion.

Apportioned effort pointing at a discipline rather than an activity. It stops being a formula and becomes an opinion, and it drifts.

Milestone credits that do not total 100 per cent. The activity can never reach complete, or it reaches complete before it is. The validation formula in §3.2 exists because this is common and invisible.

Credit released without evidence. Once a claim can be made on assertion, the sheet stops being a measurement instrument and becomes a negotiation. The evidence reference column is the whole control.

Changing a rule of credit mid-project. Reported progress moves without any work being done. Treat it as a measurement basis change under [TPL-04](#) and disclose the effect for one period.

Capping a units-complete row at 100 per cent. Quantity growth is information: it means the take-off was wrong or the scope has grown, and either way somebody needs to know. Flag it; do not hide it.

One person claiming and verifying. The two-name requirement in column R is not bureaucracy. It is the only structural control on a number that determines both reported performance and, on many projects, payment.

Backwards movement passed over in silence. Progress that reduces is either an error being corrected or a claim being withdrawn. Both need an explanation in the same period, not a quiet re-statement.

— 6. Adapting it

Safe to change. Weighting by cost instead of hours, provided you use one basis consistently — a sheet that weights some activities by hours and others by cost produces a roll-up with no meaning. Adding columns for subcontractor claim reference, payment application line, or physical location. Splitting the sheet by control account on a large project. Adding techniques that are variants of those here, such as fixed-formula start/finish rules, which are incremental milestone with two steps.

Change with care. Moving an activity from one technique to another mid-project. It is sometimes right — a planning package decomposed into measurable work should move from level of effort to units complete — but it changes reported progress without work being done, so it is a measurement basis change and belongs in the change register.

Do not remove. The evidence reference, the two-name claim and verification, the credit-total validation, and the budget-weighted roll-up. Those four are the reason the output can be defended.

Where the client's own progress measurement differs, run both and reconcile them on the sheet as an additional column pair. Do not adopt a single method that satisfies neither; the reconciliation itself is usually the most valuable thing on the page.

— 7. Completion checklist

- [] Every activity has a technique recorded in the dictionary before its first claim
- [] Every milestone activity references an approved rule of credit
- [] Every rule of credit totals exactly 100 per cent and returns [OK](#) on the validation
- [] Every credit step names the objective evidence and who releases it
- [] Level of effort is capped, and its share of each control account is known and stated
- [] Every apportioned-effort row names one base activity by identifier
- [] Percentages stored as decimal fractions in percentage-formatted cells
- [] Earned quantity equals quantity to date on every units-complete row
- [] Exception flags cleared or explained: over 100 per cent, negative movement, missing rule
- [] Roll-up computed as earned hours ÷ budgeted hours, not as an average of percentages
- [] Every row carries an evidence reference and two names
- [] Period movement reviewed activity by activity before the sheet is issued
- [] Resulting earned value reconciled to the figure entered in the earned value sheet

— Related

- [TPL-01 – Project controls execution plan](#) — where permitted techniques and the cut-off calendar are set
- [TPL-02 – Work breakdown structure and WBS dictionary](#) — where an element's technique is first recorded
- [TPL-06 – Monthly project controls report](#) — where period movement is reported and explained
- [TPL-07 – Earned value calculation sheet](#) — what the earned value produced here is used for

- [BPG-06 – Progress measurement and rules of credit](#) — the design reasoning behind technique selection
- [BPG-08 – Earned value in practice](#) — how a measured per cent complete becomes a defensible index

— Sources and standards

- PCI Master Formula Sheet ([docs/downloads/master-formula-sheet.md](#)), August 2026: the earned value identity used at §4.1 to convert a rolled-up per cent complete into earned value. Published under the credential's retired code; the credential is PCL-AI.

The five techniques named here are common to established cost engineering practice and are described in the Institute's own words. No third-party sheet, rule set or wording is reproduced; the rules of credit in §4.2 are illustrative and original.

— Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.